

# Bernardo Rivas

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Department of Mathematics and Statistics

University of Toledo

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## Research Interests

Rigorous study of nonlinear dynamical systems via computational and combinatorial topology, with emphasis on Conley index theory and topological dynamics. Current directions include data-driven and hybrid dynamical systems, with applications to systems biology, robotics and control.

## Employment

### University of Toledo

Postdoctoral Fellow, Department of Mathematics and Statistics

Supported by AFOSR MURI FA9550-23-1-0400 (PI: T. Lee, George Washington University; co-PI: W. Kalies), in collaboration with the University of Michigan

Ohio, US

2025 – Present

## Education

### Rutgers University

PhD in Mathematics, Advisor: Konstantin Mischaikow

New Jersey, US

2019 – 2025

### Universidade de São Paulo

MS and BS in Mathematics, MS Advisor: Marcio Gameiro

São Paulo, Brazil

2012 – 2019

## Preprints

### Characterizing High-dimensional Dynamics by Combinatorial-Topological Methods on a Latent Space

2026

P. Bailon, M. Gameiro, B. Gelb, W. Kalies, M. Kramar, K. Mischaikow, **B. Rivas**, E. Vieira

[arXiv:2609.01509](#) [↗](#)

### Conley Index Theory for Hybrid Systems

2026

**B. Rivas**, W. Kalies

[arXiv:2606.18501](#) [↗](#)

### Boolean Models Coarsely Sample Continuous Dynamics of Regulatory Networks

2026

B. Cummins, M. Gameiro, T. Gedeon, K. Mischaikow, **B. Rivas**

[arXiv:2606.14925](#) [↗](#)

### Global Dynamics of Ordinary Differential Equations: Wall Labelings, Conley Complexes, and Ramp Systems

2024

M. Gameiro, T. Gedeon, H. Kokubu, K. Mischaikow, H. Oka, **B. Rivas**, E. Vieira, D. Gameiro

[arXiv:2412.11078](#) [↗](#)

## Publications

### Topological Dynamics via Learned Hybrid Systems

2026

**B. Rivas**, W. Kalies, K. Iwasaki, A. Bloch, M. Ghaffari

Proceedings of the 8th Annual Learning for Dynamics and Control Conference (L4DC),

[PMLR 331:1663–1674](#) [↗](#)

## In Preparation

### Learning Cycling Signatures of Hybrid Systems

D. Hien, K. Iwasaki, **B. Rivas**

### Hybrid Attractor Lattices

W. Kalies, **B. Rivas**, T. Wehbe

### Topological Validation of Parameter Inference in Gene Regulatory Networks

W. El Khateeb, **B. Rivas**, W. Kalies

## Towards an Algorithmic Approach to Control of Dynamic Phenotypes

M. Gameiro, W. Kalies, K. Mischaikow, **B. Rivas**

### Talks

|                                                                              |                     |
|------------------------------------------------------------------------------|---------------------|
| <b>Homological Dynamics</b>                                                  | New Brunswick, NJ   |
| Conference on Algebra, K-theory and Applications                             | 2026                |
| <b>Computational Dynamics</b>                                                | Toledo, OH          |
| UToledo Mathematics Colloquium                                               | 2026                |
| <b>Computational Conley Theory</b>                                           | Vitória, Brazil     |
| UFES – Universidade Federal do Espírito Santo                                | 2026                |
| <b>Computational dynamics of gene regulatory networks</b>                    | Athens, OH          |
| Dynamics Seminar                                                             | 2025                |
| <b>Network dynamics via combinatorics, topology and homology</b>             | Toledo, OH          |
| AMS Seminar                                                                  | 2025                |
| <b>Analytical bounds for combinatorial topological dynamics</b>              | Montreal, Canada    |
| Workshop – Combinatorial topological framework for nonlinear dynamics        | 2024                |
| <b>Combinatorial topological method for gene regulatory networks</b>         | Leiden, Netherlands |
| Workshop – Computing combinatorial dynamics in high dim. biological networks | 2023                |
| <b>A computational topology approach to gene regulation</b>                  | Austin, TX          |
| Texas-Louisiana Section of SIAM 8th Annual Meeting                           | 2025                |
| <b>Global dynamics in systems biology</b>                                    | Denver, CO          |
| SIAM Dynamical Systems                                                       | 2025                |
| <b>A computational framework for dynamics of genetic regulatory networks</b> | Washington, DC, US  |
| AMS Spring Eastern Sectional Meeting                                         | 2024                |
| <b>Rigorous numerics of invariant manifolds</b>                              | São Carlos, Brazil  |
| Summer Meeting in Differential Equations                                     | 2019                |

### Mentoring

Informal research mentoring (not thesis advisor of record):

|                                                                            |                      |
|----------------------------------------------------------------------------|----------------------|
| <b>Wael El-Khateeb</b> (now Assistant Professor, Emporia State University) | University of Toledo |
| Topological validation of parameter inference in gene regulatory networks. | 2025 – 2026          |
| <b>Tony Wehbe</b>                                                          | University of Toledo |
| Computational Conley theory for hybrid systems.                            | 2025 – Present       |

### Software

|                                                                                                            |            |
|------------------------------------------------------------------------------------------------------------|------------|
| <b>MorseGraph-L4DC</b> <a href="#">↗</a> : Convex optimization and topological analysis for hybrid systems | Python     |
| <b>LatentDynamics</b> <a href="#">↗</a> : Topological analysis of learned dynamics on latent spaces        | Python     |
| <b>DSGRN</b> : Dynamic Signatures Generated by Regulatory Networks (core algorithms)                       | C++/Python |
| <b>CMGDB</b> : Conley-Morse Graph Database                                                                 | C++/Python |

### Teaching

#### Instructor of Record

|                                                      |                      |             |
|------------------------------------------------------|----------------------|-------------|
| DATA4980 – Advanced Topics in Data Science           | University of Toledo | Spring 2026 |
| 640:311 – Introduction to Real Analysis I            | Rutgers University   | Summer 2024 |
| 640:300 – Introduction to Mathematical Reasoning     | Rutgers University   | Summer 2023 |
| 640:252 – Elementary Differential Equations (online) | Rutgers University   | Summer 2022 |
| Minicourse on Differential Equations                 | ICMC-USP             | Winter 2019 |

**Teaching Assistant (Rutgers University)**

|                                                              |             |
|--------------------------------------------------------------|-------------|
| 640:311 – Introduction to Real Analysis I                    | Spring 2024 |
| 640:311 – Introduction to Real Analysis I                    | Fall 2023   |
| 640:244 – Differential Equations for Engineering and Physics | Spring 2023 |
| 640:481 – Mathematical Theory of Statistics                  | Fall 2022   |
| 640:421 – Advanced Calculus for Engineering                  | Fall 2022   |
| 640:244 – Differential Equations for Engineering and Physics | Spring 2022 |
| 640:135 – Calculus I for the Life and Social Sciences        | Fall 2021   |

**Poster Presentations**

|                                                                        |      |
|------------------------------------------------------------------------|------|
| L4DC: Learning for Dynamics and Control, Los Angeles, CA, USA          | 2026 |
| CBMS: Applied Mathematics and Machine Learning, Houston, TX, USA       | 2025 |
| Workshop of Thesis and Dissertations, ICMC-USP, Brazil                 | 2018 |
| Algebraic Topology in Data and Dynamics, Montana State University, USA | 2018 |

**Conference & Workshop Participation**

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|------------------------------------------------------------------------|------|
| Id. of Gene Regulatory Network Function from Data, Providence, RI, USA | 2022 |
| International Congress in Mathematics, Rio de Janeiro, Brazil          | 2018 |
| Summer Meeting in Differential Equations, ICMC-USP, Brazil             | 2017 |
| GeoGebra Global Gathering, Linz, Austria                               | 2015 |

**Service**

|                                                                     |             |
|---------------------------------------------------------------------|-------------|
| Referee, <i>Nonlinear Analysis: Hybrid Systems</i>                  | 2026        |
| Mentor, Directed Reading Program, Rutgers University                | 2024        |
| Member, Graduate Student Liaison Committee, Rutgers University      | 2022 – 2025 |
| Head Proctor, Rutgers University                                    | 2021 – 2023 |
| Undergraduate representative, Mathematics Department Committee, USP | 2015 – 2016 |

**Fellowships**

|                                                                              |             |
|------------------------------------------------------------------------------|-------------|
| <b>Research Internship Abroad</b> , FAPESP – Rutgers University              | 2018        |
| <b>Master's Fellowship</b> , FAPESP – Universidade de São Paulo              | 2017 – 2019 |
| <b>Science Without Borders</b> , CNPq – Queen Mary University of London      | 2014 – 2015 |
| <b>Scientific Initiation Fellowship</b> , FAPESP – Universidade de São Paulo | 2013 – 2016 |